Algebra 1 | Name Answer Key _____________
Unit 8 | Date____________________
Lesson 5 Practice | Period____________________

Re-write each expression into vertex form by completing the square.

1. $x^2 - 10x + 1$
$[x^2 - 10x + 25] + 1 - 25$
$(x-5)(x-5) - 24$
$(x-5)^2 - 24$

2. $m^2 + 8m - 11$
$[m^2 + 8m + 16] - 11 - 16$
$(m+4)(m+4) - 27$
$(m+4)^2 - 27$

3. $x^2 - 6x + 9$
$x^2 - 6x + 9$
$(x-3)(x-3)$
$(x-3)^2$

4. $j^2 - 2j - 3$
$[j^2 - 2j + 1] - 3 - 1$
$(j-1)(j-1) - 4$
$(j-1)^2 - 4$

5. $x^2 + 8x - 10$
$[x^2 + 8x + 16] - 10 - 16$
$(x+4)(x+4) - 26$
$(x+4)^2 - 26$

6. $h^2 + 2h - 3$
$[h^2 + 2h + 1] - 3 - 1$
$(h+1)(h+1) - 4$
$(h+1)^2 - 4$

7. $q^2 - 12q + 26$

$[q^2 - 12q + 36] + 26 - 36$

$(q - 6)(q - 6) - 10$

$(q - 6)^2 - 10$

8. $d^2 - 2d - 8$

$[d^2 - 2d + 1] - 8 - 1$

$(d - 1)(d - 1) - 9$

$(d - 1)^2 - 9$

9. $y^2 + 12x + 20$

$[y^2 + 12x + 36] + 20 - 36$

$(y + 6)(y + 6) - 16$

$(y + 6)^2 - 16$

10. $u^2 + 16u - 22$

$[u^2 + 16u + 64] - 22 - 64$

$(u + 8)(u + 8) - 86$

$(u + 8)^2 - 86$